

STARTUP CRM PLATFORM

Enterprise Microservices Architecture Documentation

1. Executive Summary

The Startup CRM Platform is a multi-tenant, cloud-ready, enterprise-grade SaaS system designed to manage client onboarding, HR management, employee lifecycle, secure authentication (JWT + Refresh Tokens), event-driven notifications, distributed caching, and audit tracking.

2. Architectural Vision

Design Goals: • Multi-tenant SaaS architecture • Horizontally scalable • Secure by design (Zero Trust) • Cloud-native • Event-driven communication • Production-grade observability

3. Core Architecture Patterns

• Microservices Architecture • API Gateway Pattern • Database per Service • Event-Driven Architecture (Kafka) • Distributed Caching (Redis) • Role-Based Access Control (RBAC) • Tenant Isolation Strategy

4. Microservices Breakdown

API Gateway – Routing, JWT validation, rate limiting Auth Service – Authentication, token management User Service – Profile & role management Client Service – Company onboarding & tenant management Employee Service – Employee CRUD & pagination Notification Service – Async email via Kafka File Service – Secure S3-based uploads Audit Service – Compliance logging

5. Authentication & Security

• JWT-based authentication • Refresh token stored in Redis • BCrypt password encryption • Role-based authorization • Tenant-aware filtering • HTTPS enforced • Account lockout policy

6. Multi-Tenancy Design

Shared database strategy using tenant_id column in every table. JWT carries tenant information. All queries filtered by tenant_id to ensure isolation.

7. Caching Strategy

Redis used for: • Refresh token storage • Rate limiting • Frequently accessed employee data • OTP and temporary storage

8. Database Architecture

Each service owns its database: • auth_db • client_db • employee_db • audit_db Optimizations: • Indexing on tenant_id • Pagination • Connection pooling (HikariCP) • Read replicas

9. Resilience & Fault Tolerance

• Circuit Breaker (Resilience4j) • Retry mechanism • Timeout handling • Fallback strategies • Rate limiting

10. Observability & Monitoring

• Centralized logging (ELK Stack) • Prometheus metrics • Grafana dashboards • Distributed tracing support

11. DevOps & Deployment

• Docker containerization • Kubernetes orchestration • Horizontal Pod Autoscaler • ConfigMaps & Secrets • CI/CD automation pipeline

12. Scalability Strategy

Frontend – CDN deployment Gateway – Horizontal scaling Microservices – Auto-scaling pods
Database – Read replicas Cache – Redis cluster

13. Conclusion

The Startup CRM Platform demonstrates enterprise-grade system design, advanced microservices architecture, secure authentication, distributed caching, event-driven processing, and cloud-ready scalability.